Danila Leontyev, PhD

Senior In Vivo Scientist | Preclinical Study Lead | Translational Pharmacology

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PROFESSIONAL SUMMARY

Accomplished PhD In Vivo Scientist and Preclinical Study Lead with 17+ years of hands-on experience in translational biomedical research, in vivo pharmacology, infectious disease and vaccine testing, and preclinical drug development. Proven track record functioning as a Study Director and Study Lead designing, validating, and executing small rodent (mice, rats) and humanized mouse models across infectious diseases, vaccines, gene therapies, immunology, and oncology within BSL-2 and BSL-3 environments. Extensive expertise in full-lifecycle IND-enabling programs, pharmacokinetic/pharmacodynamic (PK/PD) profiling, therapeutic efficacy, viral challenge and immunogenicity studies, biodistribution, and external CRO management under FDA, EMA, and ICH guidelines.

CORE COMPETENCIES & TECHNICAL SKILLS

- **Infectious Disease & Vaccine Models:** In vivo viral challenge, vaccine potency, and immunogenicity testing for HIV-1, Dengue virus (DENV), SARS-CoV-2, and Influenza; AAV-mediated gene therapy; BSL-3 biocontainment operations.
- **In Vivo Pharmacology & Disease Models:** Experimental study design and statistical powering; PK/PD, efficacy, biodistribution, and toxicology studies; humanized mouse platforms (CD34+ stem cell engraftment, NSG); syngeneic, xenograft, orthotopic, and subcutaneous oncology models; autoimmune and hematology models (idiopathic thrombocytopenia [ITP], rheumatoid arthritis [RA]).
- **Hands-On In Vivo Procedures:** Small rodent handling (mice, rats), multi-route dosing (IV, IP, SC, oral gavage [PO]), surgical procedures, in-life clinical monitoring, blood sampling, tissue harvesting, necropsy, and transgenic mouse colony management (breeding, genotyping, phenotyping).
- **Ex Vivo Assays & Biomarker Analysis:** Multicolor Flow Cytometry and FACS (Bio-Rad ZE5, BD LSR II, BD LSRFortessa, FlowJo), ELISA, Luminex multiplex cytokine profiling, qPCR/RT-qPCR, Western blotting, biochemical assays, and viral neutralization/inhibition assays (TZM-bl luminescence and fluorescence, p24 ELISA).
- **Molecular & Cellular Biology:** Mammalian cell culture (Jurkat, HeLa, TZM-bl, stem cells), human PBMC isolation, human CD34+ stem cell isolation and transplantation protocols, DNA/RNA extraction, PCR, and SDS-PAGE.
- **Study Direction & Regulatory Compliance:** IND-enabling study coordination, external CRO and vendor management, authoring study protocols, SOPs, and study reports aligned with FDA, EMA, and ICH guidelines; GraphPad Prism and statistical analysis software.

PROFESSIONAL EXPERIENCE

LABORATORY FOR GENE THERAPY | August 2020 – Present

Senior In Vivo Scientist & Preclinical Study Lead (Expert, Senior Research Associate)

- Direct preclinical in vivo development of AAV-based gene therapies for HIV-1 as Study Lead, managing the full study lifecycle from protocol design and statistical powering to hands-on execution in BSL-3 containment.
- Design and execute in vivo pharmacology, viral challenge, and vaccine efficacy/potency studies in murine models across HIV-1, Dengue virus (DENV), and Influenza to evaluate therapeutic efficacy, immunogenicity, PK/PD, and biodistribution.
- Develop, optimize, and validate humanized mouse models for preclinical testing of gene therapies and vaccines, enabling quantitative evaluation of antiviral and immunological responses via flow cytometry, ELISA, and qPCR.
- Lead preclinical collaborations with international academic partners and external CROs, ensuring alignment on study design, timelines, regulatory compliance, and data integrity.
- Provide technical leadership, mentoring, and hands-on training to research staff in small rodent experimental procedures, study design, and regulatory compliance.
- Provided core scientific expertise for the design and construction of a new BSL-3 animal facility, establishing operational SOPs and biocontainment standards for advanced in vivo infectious disease research.

BIOCAD (Biotechnology & Pharmaceutical) | September 2018 – July 2020

Senior Research Scientist / In Vivo Preclinical Lead (Head of the Laboratory for In Vivo Preclinical Research)

- Functioned as Study Director and Lead Scientist directing the development, optimization, validation, and execution of preclinical in vivo pharmacodynamic (PD) and efficacy models in small rodents supporting drug discovery in infectious diseases, oncology, autoimmune diseases, and genetic disorders (SMA, Hemophilia IX).
- Oversaw preclinical research programs in vaccine development (including SARS-CoV-2) and AAV gene therapies (SMA, Hemophilia IX), generating efficacy and safety data that supported preclinical-to-clinical transitions across multiple therapeutic pipelines.
- Established and advanced solid tumor (subcutaneous, orthotopic) and hematological malignancy models across syngeneic, xenograft, and humanized mouse platforms to evaluate novel cancer and immuno-oncology therapeutics.
- Designed and conducted nonclinical in vivo studies under EMA, FDA, and ICH guidelines, managing concurrent study timelines and cross-functional workflows for IND-enabling packages.
- Planned, managed, and reviewed outsourced preclinical studies with external CROs and vendors, ensuring study design rigor, regulatory alignment, and high-quality data for regulatory submissions.
- Mentored and trained scientific staff and laboratory technicians, elevating team capabilities in in vivo pharmacology, animal model development, and regulatory requirements.

ST. MICHAEL'S HOSPITAL | Toronto, ON, Canada | June 2017 – June 2018

Senior Research Associate (In Vivo Pharmacology)

- Developed and optimized translational in vivo murine models of autoimmunity and hematology, including idiopathic thrombocytopenia (ITP) and rheumatoid arthritis (RA), to support preclinical evaluation of novel immunomodulatory therapeutics.
- Independently designed, executed, and analyzed pharmacokinetic (PK), pharmacodynamic (PD), and mechanism-of-action studies, generating efficacy data that informed candidate selection and development strategy.
- Collaborated with multidisciplinary internal and external research teams, authoring study reports and presenting findings to drive data-driven translational decision-making.

THE HOSPITAL FOR SICK CHILDREN (SickKids) | Toronto, ON, Canada | November 2016 – May 2017

Research Fellow (Preclinical Antiviral & Gene Therapy)

- Led preclinical screening of anti-HIV drug candidates and AAV vector testing under BSL-3 biocontainment conditions, contributing to the advancement of the institutional antiviral development pipeline.
- Developed and implemented PK/PD research protocols, performed viral inhibition and neutralization assays (TZM-bl luminescence/fluorescence assay, p24 ELISA), and authored detailed study reports for regulatory readiness.
- Delivered comprehensive data analysis, technical documentation, and scientific presentations to internal teams and collaborators to support regulatory submissions and manuscripts.

UNIVERSITY OF TORONTO | Toronto, ON, Canada | July 2013 – October 2016

Postdoctoral Research Associate (In Vivo HIV & Vaccine Models)

- Developed and managed a collaborative inter-institutional UHN–UofT preclinical project evaluating HIV-1 therapeutics and vaccine candidates in vivo, coordinating cross-functional teams and study execution.
- Designed, validated, and applied humanized mouse models (NSG) in a BSL-3 facility as core translational platforms for preclinical HIV drug and vaccine development.
- Executed advanced in vivo and ex vivo experiments: human CD34+ stem cell isolation and transplantation, mouse humanization, multi-route dosing, blood/tissue harvesting, multicolor flow cytometric engraftment analysis, and HIV-1 challenge studies.
- Authored institutional SOPs and implemented biosafety compliance protocols; trained and supervised students and technical staff in animal handling, sample collection, and data analysis.

CANADIAN BLOOD SERVICES | Toronto, ON, Canada | February 2008 – October 2016

Research Associate (Jul 2011 – Oct 2016) & Postdoctoral Research Fellow (Feb 2008 – Feb 2011)

- Directed preclinical in vivo pharmacology, hematology, and immunology projects investigating therapeutic mechanisms across autoimmunity, hematology, and infectious disease models.
- Developed, validated, and optimized murine models of immune/idiopathic thrombocytopenia (ITP), rheumatoid arthritis (RA), and HIV infection to support translational drug evaluation.
- Designed and executed pharmacodynamic (PD) and mechanistic studies that uncovered cytokine-mediated pathways in intravenous immunoglobulin (IVIg) therapy using PBMC isolation, DNA/RNA extraction, qPCR, and multiplex plasma cytokine profiling (Luminex, ELISA).
- Managed and maintained transgenic mouse colonies, overseeing breeding, genotyping, and phenotyping to ensure continuous availability of validated preclinical models.

EDUCATION

Doctor of Philosophy (PhD) in Physiology & Biology | 2005
Saratov State University Named after N.G. Chernyshevsky

SELECTED PEER-REVIEWED PUBLICATIONS

- **Leontyev DS**, Urusov FA, Glazkova DV, et al. *Triple rAAV9 Vector Combinations Encoding Broadly Neutralizing Antibodies Effectively Suppress HIV-1 Infection in Humanized Mice.* Int J Mol Sci. 2025;26(22):11051. (PMID: 41303534)
- *Targeting ABL1 or ARG Tyrosine Kinases to Restrict HIV-1 Infection in Primary CD4+ T-Cells or in Humanized NSG Mice.*
- *Cytokine profiles in mouse models of experimental immune thrombocytopenia reveal a lack of inflammation and differences in response to intravenous immunoglobulin depending on the mouse strain.*
- Complete bibliography available at: ncbi.nlm.nih.gov/myncbi/danila.leontyev.1/bibliography/public/